

Higher-Order Asymptotics of the Discrepancy Function under Fixed Model and Distributional Misspecification

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Purpose

The companion note `rmsea_asymptotics.tex` fixes the meaning of RMSEA by a Pitman drift: the misspecification is sent to zero at the $\sqrt{n}$ rate so the test statistic converges to a (noncentral) χ^2 and the noise inflation is the first-order Satorra–Bentler trace $\text{tr}(UT)$. This note drops the drift. It expands the fitted discrepancy $\hat{F}$ to order n^{-1} under *fixed* misspecification in both the covariance structure (σ_0 off the model) and the distribution ($\Gamma \neq$ normal theory), for a general minimum-discrepancy estimator. The point is to see what the local sequence silently discards and to ask whether GLS-RMSEA differs from normal-theory RMSEA at all once the drift is removed.

The single result is the bias expansion

$$\mathbb{E}[\hat{F}] = F_0 + \frac{1}{n} \left[\underbrace{\text{tr}(\tilde{U}\Gamma)}_{\text{generalized } df} + \underbrace{\beta_W}_{\text{weight influence, } \propto e} \right] + o(n^{-1}), \quad (1)$$

with two objects that the drift kills. The trace carries the *curvature-corrected* residual-weight matrix $\tilde{U} = V - V\dot{\sigma}B^{-1}\dot{\sigma}'V$, built from the population *Hessian* $B = A - K$ of the discrepancy rather than its Gauss–Newton part $A = \dot{\sigma}'V\dot{\sigma}$; the two agree only when the residual–curvature coupling K vanishes (correct specification, or a model linear in θ). The remainder β_W is the influence of an estimated weight; it is identically zero for a fixed weight (ULS) and vanishes as the residual $e \rightarrow 0$ for any estimator. Three corollaries read off (1): a curvature-corrected, robust bias correction for RMSEA

(Corollary 1); an honest Wald confidence interval that needs no noncentral- χ^2 inversion (Corollary 3); and the ML/GLS/ULS comparison (Section 7).

Two structural remarks orient the rest. First, $\tilde{U}$ uses the Hessian B where the first-order projector U uses A ; this is the fit-index sibling of the observed-Hessian bread that replaces Fisher information in the misspecification-robust standard errors of `misspec_observed_bread.tex`, and β_W is the fit-index sibling of the data-dependent-weight meat term there. Both corrections are zero at H_0 and leading-order off it; the order-promotion is the same theorem twice. Second, in the language of `aic_tic_rmsea_bias.tex`, $\text{tr}(\tilde{U}\Gamma)$ is to $\text{tr}(U\Gamma)$ as TIC's $\text{tr}(I^{-1}J)$ is to AIC's k , and then one order deeper: the curvature term is a correction past even the first-order robust (TIC-like) trace.

1 Setup

Stack the $p^* = p(p+1)/2$ distinct sample covariances as $s = \text{vech } \hat{\Sigma}$ (append means for a mean structure; the algebra is unchanged), with population limit $s \xrightarrow{p} \sigma_0$ and

$$z := \sqrt{n}(s - \sigma_0) \xrightarrow{d} N(0, \Gamma), \quad (2)$$

Γ the asymptotic covariance of the moments, normal-theory or not. The model is a smooth map $\theta \mapsto \sigma(\theta)$, $\theta \in \mathbb{R}^t$, $df = p^* - t$. A minimum-discrepancy estimator minimizes a quadratic form in the residual $r(\theta) = s - \sigma(\theta)$,

$$\hat{F} = \min_{\theta} r(\theta)' \hat{V} r(\theta), \quad \hat{V} \xrightarrow{p} V, \quad (3)$$

with weight $\hat{V}$ either fixed (ULS) or a smooth function of the data (GLS). Maximum likelihood is not exactly of this form; it is treated as the reweighted limit in Section 7.

Pseudo-true value and population residual. Let

$$\theta_* = \arg \min_{\theta} (\sigma_0 - \sigma(\theta))' V (\sigma_0 - \sigma(\theta)), \quad e := \sigma_0 - \sigma(\theta_*), \quad F_0 = e' V e. \quad (4)$$

Write $\dot{\sigma} = \partial\sigma/\partial\theta'$ at θ_* ($p^* \times t$, full column rank) and the model Hessian tensor $\ddot{\sigma}$, $(\ddot{\sigma})_{a,ij} = \partial^2 \sigma_a / \partial \theta_i \partial \theta_j$. The first-order condition for (4) is the V -orthogonality of the residual to the tangent space,

$$\dot{\sigma}' V e = 0. \quad (5)$$

Under correct specification $e = 0$; under fixed misspecification $e = O(1)$ and (5) is all that survives of it.

Curvature objects. Define the Gauss–Newton matrix, the residual–curvature coupling, and the population Hessian of the discrepancy,

$$A := \dot{\sigma}' V \dot{\sigma}, \quad K_{ij} := (Ve)'(\ddot{\sigma})_{:,ij} = \sum_a (Ve)_a(\ddot{\sigma})_{a,ij}, \quad B := A - K = \frac{1}{2} \partial_{\theta\theta}^2 (e'Ve). \quad (6)$$

K is the only object in (6) that requires both a nonzero residual (Ve) and a curved model ($\ddot{\sigma} \neq 0$); it is the misspecification signature. With B in hand define the standard residual projector and its curvature-corrected analogue

$$U := V - V \dot{\sigma} A^{-1} \dot{\sigma}' V, \quad \tilde{U} := V - V \dot{\sigma} B^{-1} \dot{\sigma}' V = V - V P V, \quad P := \dot{\sigma} B^{-1} \dot{\sigma}'. \quad (7)$$

U is a genuine residual projector ($U \dot{\sigma} = 0$, idempotent in the V -metric); $\tilde{U}$ is a perturbation of it that no longer annihilates the tangent ($\tilde{U} \dot{\sigma} = -V \dot{\sigma} B^{-1} K$), reducing to U as $K \rightarrow 0$. U is the object of every limit law in the drift regime (`rmsea_asymptotics.tex`); $\tilde{U}$ is its fixed-misspecification replacement.

Weight fluctuation. For an estimated weight expand

$$\hat{V} = V + n^{-1/2} \dot{V} + \frac{1}{2} n^{-1} \ddot{V} + o_p(n^{-1}), \quad (8)$$

where $\dot{V} = O_p(1)$ is linear in z (the derivative of the weight map along the moment fluctuation) and $\ddot{V} = O_p(1)$ is the second-order term. Write the residual-projected weight fluctuation

$$w := \dot{V} e, \quad (9)$$

an $O_p(1)$ vector that is identically zero for a fixed weight and is $O(\|e\|)$ otherwise. For GLS, $\dot{V}$ is driven by the raw moment fluctuation z ; for ML, by the projected fluctuation $\dot{\sigma} (\hat{\theta} - \theta_*)$ (Section 7).

2 The stochastic expansion

Theorem 1 (Second-order expansion of $\hat{F}$). *Assume (2), identification ($\dot{\sigma}$ full rank, B nonsingular), a weight that does not depend on θ (ULS, GLS, or any fixed/data weight), and the usual smoothness. Then*

$$\hat{F} = F_0 + n^{-1/2} (2e'Vz + e'\dot{V}e) + n^{-1} \Phi + o_p(n^{-1}), \quad (10)$$

with

$$\Phi = z' \tilde{U} z + 2w'(I - PV)z - w'Pw + \frac{1}{2}e'\ddot{V}e. \quad (11)$$

Proof. Write $u = \hat{\theta} - \theta_* = O_p(n^{-1/2})$ and split it as $u = n^{-1/2}\xi + O_p(n^{-1})$. Expand the residual at the estimator,

$$r(\hat{\theta}) = e + \rho_1 + \rho_2 + o_p(n^{-1}), \quad \rho_1 = n^{-1/2}z - \dot{\sigma}u, \quad \rho_2 = -\frac{1}{2}(\ddot{\sigma})[u, u],$$

where $(\ddot{\sigma})[u, u]$ has a -component $u'(\ddot{\sigma})_{:,a}u$. The estimating equation $\dot{\sigma}(\hat{\theta})'\hat{V}r(\hat{\theta}) = 0$, expanded to $O_p(n^{-1/2})$ with $\dot{\sigma}(\hat{\theta}) = \dot{\sigma} + (\ddot{\sigma})[u, \cdot]$ and (8), gives after using (5)

$$-Au + (\ddot{\sigma})[u, \cdot]'Ve + n^{-1/2}\dot{\sigma}'Vz + n^{-1/2}\dot{\sigma}'\dot{V}e = o_p(n^{-1/2}).$$

The second term is Ku by (6); collecting,

$$(A - K)u = n^{-1/2}\dot{\sigma}'(Vz + \dot{V}e) + o_p(n^{-1/2}) \implies \xi = B^{-1}\dot{\sigma}'(Vz + w), \quad u = n^{-1/2}\xi. \quad (12)$$

The weight fluctuation enters ξ only through $w = \dot{V}e$, hence only under misspecification. Now substitute into $\hat{F} = (e + \rho)'\hat{V}(e + \rho)$, $\rho = \rho_1 + \rho_2$, and use (5) twice. The cross term with ρ_1 collapses to the linear part, $2e'V\rho_1 = 2n^{-1/2}e'Vz$ (the $\dot{\sigma}u_1$ piece dies by (5)); the cross term with ρ_2 kills the second-order parameter increment and leaves only curvature, $2e'V\rho_2 = -u'Ku$; and $e'\dot{V}e$ contributes the weight terms $n^{-1/2}e'\dot{V}e + \frac{1}{2}n^{-1}e'\ddot{V}e$. Together with $\rho_1'V\rho_1 = n^{-1}(z'Vz - 2\xi'\dot{\sigma}'Vz + \xi' A\xi)$ and the $\dot{V}\rho_1$ cross term $2n^{-1}w'(z - \dot{\sigma}\xi)$, the $O(n^{-1})$ part is

$$\Phi = z'Vz - 2\xi'\dot{\sigma}'Vz + \xi'B\xi + 2w'(z - \dot{\sigma}\xi) + \frac{1}{2}e'\ddot{V}e,$$

where $\xi' A\xi - u'Ku/n^{-1}$ combined to $\xi'B\xi$. Insert $\xi = B^{-1}\dot{\sigma}'(Vz + w)$. With $P = \dot{\sigma}B^{-1}\dot{\sigma}'$ one gets $\xi'B\xi = (Vz + w)'P(Vz + w)$, $\xi'\dot{\sigma}'Vz = (Vz + w)'PVz$ and $\dot{\sigma}\xi = P(Vz + w)$. Expanding and cancelling,

$$\Phi = z'(V - VPV)z + 2w'(I - PV)z - w'Pw + \frac{1}{2}e'\ddot{V}e,$$

which is (11) with $\tilde{U} = V - VPV$. $\square$

Remark (The cancellation is the content). Everything hinges on the population FOC (5). It removes the tangent-direction sampling noise from the linear term (leaving $2e'Vz = 2e'Uz$, since $e'V = e'U$ by (5)) and removes the second-order parameter increment u_2 from the bias, so the bias needs ξ only to first order. What remains is genuinely a residual-space object, projected by $\tilde{U}$.

3 Bias and the curvature-corrected df

Corollary 1 (Generalized degrees of freedom). *Under the conditions of Theorem 1,*

$$\mathbb{E}[\hat{F}] = F_0 + \frac{1}{n} [\text{tr}(\tilde{U}\Gamma) + \beta_W] + o(n^{-1}), \quad \beta_W = 2\mathbb{E}[w'(I - PV)z] - \mathbb{E}[w'Pw] + \frac{1}{2}e'\mathbb{E}[\ddot{V}]e. \quad (13)$$

For a fixed weight $\beta_W \equiv 0$; for any estimator $\beta_W = O(\|e\|)$, so it vanishes at H_0 . Under correct specification $e = 0$, $K = 0$, $\tilde{U} = U$ and (13) reduces to $\mathbb{E}[\hat{F}] = \text{tr}(U\Gamma)/n + o(n^{-1})$, the Satorra–Bentler mean *cdf* = $\text{tr}(U\Gamma)$ of *rmsea_asymptotics.tex*, which is *df* when $V = \Gamma^{-1}$.

Proof. Take expectations in (10). The $n^{-1/2}$ term has mean zero ($\mathbb{E}z = 0$, $\mathbb{E}\dot{V} = 0$). For the n^{-1} term, $\mathbb{E}[z'\tilde{U}z] = \text{tr}(\tilde{U}\Gamma)$ by the quadratic-form mean and the remaining terms define β_W . $\square$

The estimand correction therefore generalizes the chain

$$df \longrightarrow \text{tr}(U\Gamma) \longrightarrow \text{tr}(\tilde{U}\Gamma) + \beta_W,$$

in increasing order of fidelity: *df* assumes a correctly weighted, correctly distributed null; $\text{tr}(U\Gamma)$ allows distributional misspecification but freezes the model at its tangent; $\text{tr}(\tilde{U}\Gamma) + \beta_W$ allows fixed structural misspecification, correcting the projector for curvature and adding the weight influence. A curvature-and-weight corrected RMSEA is

$$\hat{\varepsilon}_{\text{ho}} = \sqrt{\max\left(\frac{\hat{F}}{df} - \frac{\text{tr}(\tilde{U}\hat{\Gamma}) + \hat{\beta}_W}{n df}, 0\right)}. \quad (14)$$

Remark (Curvature is real for SEM). $K \neq 0$ requires $\ddot{\sigma} \neq 0$. Factor and path models are nonlinear in θ (loadings multiply, paths compose), so $\ddot{\sigma} \neq 0$ generically and the curvature term is present whenever the model is wrong. It vanishes for models linear in θ (e.g. a free covariance matrix with linear equality constraints), where $\tilde{U} = U$ exactly even under misspecification.

4 Comparison with the first-order trace

The fixed-misspecification trace is the Satorra–Bentler trace plus an additive series, so it slots directly against earlier robust RMSEA.

Corollary 2 (Bread decomposition). *Let $N = \dot{\sigma}' V \Gamma V \dot{\sigma}$ (the parameter-space meat). Both traces are the same total noise $\text{tr}(V \Gamma)$ minus a model-absorbed part that differs only in its bread:*

$$\text{tr}(U \Gamma) = \text{tr}(V \Gamma) - \text{tr}(A^{-1} N), \quad \text{tr}(\tilde{U} \Gamma) = \text{tr}(V \Gamma) - \text{tr}(B^{-1} N). \quad (15)$$

The first-order (Satorra–Bentler) trace uses the Gauss–Newton bread A^{-1} ; the fixed-misspecification trace uses the Hessian bread B^{-1} . This is the same $A^{-1} \rightarrow B^{-1}$ swap (expected information to observed Hessian) that turns a naive standard error into a misspecification-robust one (`misspec_observed_bread.tex`), here inside the fit-index trace. Expanding B^{-1} about A^{-1} with $B = A - K$ and $B^{-1} - A^{-1} = \sum_{m \geq 1} (A^{-1} K)^m A^{-1}$ gives the additive series

$$\text{tr}(\tilde{U} \Gamma) = \underbrace{\text{tr}(U \Gamma)}_{\text{robust RMSEA}} - \text{tr}(KM) - \text{tr}(KA^{-1}KM) - \dots, \quad M := A^{-1} N A^{-1}, \quad (16)$$

geometric in $A^{-1}K$ (ratio its spectral radius). The leading correction $\text{corr}_1 = -\text{tr}(KM)$ is $O(\|e\|)$: the residual-weighted model curvature $K = (Ve)'\ddot{\sigma}$ contracted against M , the Gauss–Newton sandwich covariance of $\hat{\theta}$ (the parameter covariance read at the tangent; it equals the fixed-misspecification $\text{acov } \hat{\theta} = B^{-1} N B^{-1}$ only at $K = 0$). Earlier robust RMSEA is the $K = 0$ truncation.

Remark (Magnitudes, and ML). In the ULS check (Section 8) the baseline $\text{tr}(U \Gamma) = 3.38$ and the corrections $\text{corr}_1 = -0.55$, $\text{corr}_2 = -0.38$, $\text{corr}_3 = -0.12, \dots$ sum to $\text{tr}(\tilde{U} \Gamma) = 2.26$; corr_1 alone recovers half the gap, $\text{corr}_{1:2}$ four-fifths. For ML there are two baselines to deform from, the metric ($W_* \rightarrow V_0$) and the bread ($B \rightarrow A$), so the analogous expansion carries two families of corrections, both $O(\|e\|)$ and both vanishing under correct specification.

5 The honest interval

Corollary 3 (CLT and Wald interval, no noncentral inversion). *If $F_0 > 0$, then from (10)*

$$\sqrt{n}(\hat{F} - F_0) \xrightarrow{d} N(0, \omega^2), \quad \omega^2 = \text{Var}(2e'Vz + e'\dot{V}e) = (2Ve + v_e)'\Gamma(2Ve + v_e), \quad (17)$$

where v_e is the vector representing the linear functional $z \mapsto e'\dot{V}e = v_e'z$; for a fixed weight $v_e = 0$ and $\omega^2 = 4e'V\Gamma V e = 4e'U\Gamma U e$. By the delta method, with $\varepsilon = \sqrt{F_0/df}$,

$$\sqrt{n}(\hat{\varepsilon} - \varepsilon) \xrightarrow{d} N\left(0, \frac{\omega^2}{4df F_0}\right), \quad (18)$$

giving a Wald interval for ε directly on the discrepancy scale.

Equation (18) is the fixed-misspecification interval: under a genuinely wrong model the statistic is asymptotically normal, not noncentral χ^2 , and the interval is a plain sandwich, not a CDF inversion. It complements rather than replaces the noncentral interval of `rmsea_asymptotics.tex`: that one is built for the near-null regime ($F_0 \downarrow 0$) where (18) degenerates (ω^2/F_0 blows up), while (18) is built for clearly-misspecified models where the noncentral inversion is least trustworthy. The crossover is exactly where neither device is comfortable, which is the honest statement of where RMSEA interval coverage is hard.

6 What the drift discards

Put back the Pitman sequence $e_n = n^{-1/2}\delta$ with δ a fixed off-model direction, the regime of `rmsea_asymptotics.tex`. Then $Ve_n = O(n^{-1/2})$ so $K = O(n^{-1/2}) \rightarrow 0$, hence $B \rightarrow A$ and $\tilde{U} \rightarrow U$; the weight influence $w = \dot{V}e_n = O_p(n^{-1/2}) \rightarrow 0$, so $\beta_W \rightarrow 0$; and the normal CLT (17) degenerates while $n\hat{F}$ acquires the (noncentral) χ^2 limit. The drift annihilates precisely the two corrections this note is about.

	fixed misspecification ($e = O(1)$)	Pitman drift ($e = n^{-1/2}\delta$)
limit law of $\hat{F}$	normal, $\sqrt{n}(\hat{F} - F_0) \xrightarrow{d} N(0, \omega^2)$	$n\hat{F} \xrightarrow{d} (\text{noncentral } \chi^2 \text{ mixture})$
noise/bias mean	$\text{tr}(\tilde{U}\Gamma)$, curvature-corrected	$\text{tr}(U\Gamma) = cdf$, no curvature
weight influence β_W	$O(\ e\)$, present	$\rightarrow 0$
ML vs GLS	differ (two-metric, pseudo-true)	coincide to leading order
interval	Wald via ω^2 (18)	noncentral- χ^2 inversion

The local-asymptotics motivation is not wrong; it is a deliberate choice to work in the only regime where the df subtraction is exact and the curvature and weight terms are negligible. Outside it, those terms are the leading description of a wrong model's fit.

7 ML, GLS, ULS

The estimators differ in where the weight is anchored, and once the drift is gone this separates their RMSEAs. The unifying tool is that $\tilde{U}$ is, for *any* smooth discrepancy, half the Hessian of the profiled criterion.

Lemma 1 (Profiled-Hessian form). *For a smooth discrepancy $F(s, \theta)$ with population minimizer θ_* , Corollary 1 holds with*

$$\tilde{U} = \frac{1}{2} \nabla_s^2 \left[\min_{\theta} F(s, \theta) \right] \Big|_{s=\sigma_0} = V_{ss} - \frac{1}{4} C B^{-1} C', \quad (19)$$

where $V_{ss} = \frac{1}{2} \partial_{ss}^2 F$, $C = \partial_{s\theta}^2 F$ and $B = \frac{1}{2} \partial_{\theta\theta}^2 F$ at (σ_0, θ_*) . The fixed-weight quadratic (7) is the case $V_{ss} = V$, $C = -2V\dot{\sigma}$ (then $\frac{1}{4}CB^{-1}C' = V\dot{\sigma}B^{-1}\dot{\sigma}'V$). A non-quadratic F enters only through these three second derivatives; there is no separate higher-derivative term at $O(1/n)$, and the data skewness enters the bias only at $O(n^{-3/2})$. For a data-dependent weight $V(s)$ the s -derivatives of the weight populate V_{ss} and C , reproducing β_W of Corollary 1 in trace form.

ULS. Fixed weight, $\dot{V} = 0$, so $\beta_W \equiv 0$ and the bias is purely $\text{tr}(\tilde{U}T)/n$. ULS is never efficient, so $\text{tr}(UT) \neq df$ already at leading order regardless of normality; the curvature term is then a clean second correction with no weight bookkeeping. This is the cleanest place to exhibit $\tilde{U} \neq U$.

GLS. Weight $\hat{V} = V(s)$ at the sample moments, so $\dot{V}$ is the derivative of $V(\cdot)$ along the *raw* fluctuation z . The full β_W is present, leading-order in the residual through the cross term $2\mathbb{E}[w'(I - PV)z]$, and ω^2 in (17) acquires the v_e term.

ML. ML is not a fixed quadratic: its normal equation $\dot{\sigma}'W(\theta)(\sigma_0 - \sigma(\theta)) = 0$ anchors the weight $W(\theta) = \frac{1}{2}D'(\Sigma(\theta)^{-1} \otimes \Sigma(\theta)^{-1})D$ at the *fitted model*. Two consequences. First, the pseudo-true value solves a model-metric projection, so under fixed misspecification $\theta_*^{\text{ML}} \neq \theta_*^{\text{GLS}}$ and $F_0^{\text{ML}} \neq F_0^{\text{GLS}}$: the two estimate *different population numbers*, already at zeroth order, measuring misfit in different metrics (Olsson, Foss, Troye and Howell [2]). Second, by Lemma 1 the noise weight is the moment-Hessian $V_{ss} = V_0 = \frac{1}{2}D'(\Sigma_0^{-1} \otimes \Sigma_0^{-1})D$ at the *data*, while the cross-term $C = -2W_*\dot{\sigma}$ carries the *model* weight $W_* = \frac{1}{2}D'(\Sigma_*^{-1} \otimes \Sigma_*^{-1})D$, $\Sigma_* = \Sigma(\theta_*)$, so

$$\tilde{U}_{\text{ML}} = V_0 - W_*\dot{\sigma} B_{\text{ML}}^{-1} \dot{\sigma}' W_*. \quad (20)$$

This is the two-metric object: sampling noise weighted in the data metric, residuals projected in the model metric. The non-quadraticity of the log-det is absorbed entirely into $(V_0, W_*, B_{\text{ML}})$; there is no separate third-derivative term (correcting an earlier draft). It all collapses under correct specification ($\Sigma_* = \Sigma_0$, $W_* = V_0$, $e = 0$), where ML, GLS and ULS coincide to the order that matters. The numerical check confirms (20) to finite-difference precision.

So the answer to “does GLS-RMSEA do anything that normal-theory RMSEA does not” is yes, and sharper than expected: under fixed misspecification the two already target different estimands (different metric), and on top of that carry different $O(1/n)$ biases through β_W and B . Under the drift, all of this disappears and the classical equivalence is restored. The daylight between estimators lives entirely in the fixed-misspecification, higher-order regime.

8 Numerical check

A one-factor model is fit to a two-factor population (six indicators, three per factor, factor correlation 0.55), a misspecification the one-factor model cannot absorb ($df = 9$; ULS $F_0 = 0.23$, ML $F_0 = 0.49$, the differing estimands of Section 7). Two checks.

Deterministic. By Lemma 1 the bias coefficient is half the Hessian of the profiled criterion, which finite differencing pins exactly, with no Monte Carlo. Comparing $\frac{1}{2}\nabla_s^2[\min_{\theta} F]$ at σ_0 to the closed forms:

	$\ \frac{1}{2}H_{\text{prof}} - \tilde{U}\ /\ \tilde{U}\ $	$\text{tr}(\tilde{U}\Gamma_{\text{NT}})$ corrected	naive (Gauss–Newton A)
ULS	2×10^{-8}	2.26	3.38
ML	9×10^{-6}	1.24	9.07

ULS is a single-metric quadratic (positive control); ML is the two-metric $\tilde{U}_{\text{ML}}$ of (20), confirmed to finite-difference precision. Replacing the Hessian B by Gauss–Newton A is wrong by a factor of several, for ML by $7\times$ (and with the opposite conclusion about how much sampling noise a wrong model spends): discarding the curvature is not a small error.

Monte Carlo (ULS, $\beta_W = 0$). Confirms the Hessian-to-bias step, $N(\mathbb{E}[\hat{F}] - F_0) \rightarrow \text{tr}(\tilde{U}\Gamma)$, under normal and t_6 data; the sample covariance is the unbiased $\div(N-1)$ form (the $\div N$ form carries its own $O(1/N)$ moment bias contaminating this order), Γ_{NT} closed form under normality and Monte-Carlo under t_6 :

data	N	$N(\mathbb{E}[\hat{F}] - F_0)$ (MC se)	naive $\text{tr}(UT)$	corrected $\text{tr}(\tilde{U}T)$
normal	200	2.23 (0.05)	3.38	2.26
normal	800	2.21 (0.10)	3.38	2.26
t_6	200	4.22 (0.08)	7.00	4.74
t_6	800	4.78 (0.16)	7.00	4.74

By $N = 800$ the empirical bias is within one Monte-Carlo standard error of $\text{tr}(\tilde{U}T)$ in both conditions; the naive trace is off by the full gap (over twenty standard errors at $N = 200$, normal). ML is validated deterministically rather than by Monte Carlo: its bias has a large higher-order tail and small- N improper-solution inflation that the exact Hessian check sidesteps. The curvature term is not a third-decimal refinement; it is a third to a half of the whole bias. Script: `higher_order_discrepancy_check.py`.

9 Connections and caveats

Remark (Hessian bread, weight meat: the SE sibling). $\tilde{U}$ uses the population Hessian $B = A - K$ where U uses the Gauss–Newton A ; under correct specification $K = 0$ and they agree, the discrepancy-function analogue of the information equality. This is the same substitution that the misspecification-robust standard errors of `misspec_observed_bread.tex` make when they replace the Fisher bread by the observed Hessian, and β_W is the fit-index counterpart of their data-dependent-weight meat term. Both corrections are $O(\|e\|)$, hence zero at H_0 and leading-order off it: the order-promotion (Newey–McFadden null irrelevance, Hall–Inoue off-null promotion) is one phenomenon appearing in the variance there and in the bias here.

Remark (TIC analogy, one order deeper). `aic_tic_rmsea_bias.tex` reads df as the AIC penalty and notes the robust trace as its TIC analogue. Here $\text{tr}(UT)$ is that TIC-like trace (first order), and $\text{tr}(\tilde{U}T) + \beta_W$ is the next order. The user’s instinct stands: the df correction is AIC-like (a fixed integer), the robust trace is TIC-like (a sandwich trace), and the practical complaint about TIC, that $\text{tr}(\hat{I}^{-1}\hat{J})$ is noisy because $\hat{J}$ is a fourth-moment object, transfers verbatim: $\text{tr}(\tilde{U}\hat{\Gamma})$ inherits the sampling variance of $\hat{\Gamma}$. The higher-order expansion is what lets one judge the trade: when $\|e\|$ and the kurtosis are small the AIC-like df is nearly unbiased and lower variance; when they are not, the bias of the cheap correction is exactly (13) and may dominate. A structured (model-implied or shrunk) $\hat{\Gamma}$ is the natural middle estimator, the RMSEA analogue of a regularized TIC.

Remark (Caveats). The truncation $\max(\cdot, 0)$ in (14) re-introduces an $O(n^{-1/2})$ bias near $F_0 = 0$, where the fixed-misspecification expansion is least relevant

and the near-null device should take over. Data skewness enters the bias only at $O(n^{-3/2})$ (Lemma 1); the t_6 shortfall at $N = 200$ in the table is that tail. The expansion assumes finite fourth moments for Γ and finite sixth moments for the n^{-1} bias to have finite expectation; under heavier tails (13) is a stochastic-order statement, not a moment statement.

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